Next-Day Air Quality Index Forecasting and Health-Alert Classification for 26 Indian Cities using Classical Machine-Learning Models

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Abstract. Air pollution is one of the leading health risks in India, yet public advisories are usually issued only after a bad-air day has begun. This study asks whether classical machine-learning (ML) models taught in an undergraduate syllabus can forecast the next day's Air Quality Index (AQI) and warn of "Poor"-or-worse days (AQI > 200) one day in advance. We use the publicly available city-level daily data for 26 Indian cities (2015–2020) compiled from the Central Pollution Control Board. After per-city construction of consecutive-day pairs, median imputation, winsorisation, standardisation and one-hot encoding, 24,343 samples were split chronologically (train 2015–2018; test January 2019–February 2020; separate COVID-19 lockdown period). Eight regressors and seven classifiers were tuned with five-fold forward-chaining (time-series) cross-validation and compared with a persistence baseline. A support-vector regressor gave the lowest test RMSE (34.96; R^2 = 0.891), 17.6% below persistence, while regularised linear models were within 6%. For the health alert, logistic regression achieved F1 = 0.866 and ROC-AUC = 0.981, against F1 = 0.833 for persistence. During the 2020 lockdown, however, the tuned model was worse than persistence, showing sensitivity to sudden changes in emissions. Simple, interpretable models are therefore adequate for city-level early warning in India, provided they are re-trained when conditions change.

Keywords: air quality index; India; CPCB; regression; classification; support vector machine; logistic regression; time-series cross-validation; SDG 3; SDG 11

1 Introduction

Ambient particulate-matter pollution is a major cause of death and disease in India; the state-level Global Burden of Disease analysis attributed about 1.67 million deaths in 2019 to air pollution [1]. Under the National Clean Air Programme, the Central Pollution Control Board (CPCB) reports a daily AQI that maps pollutant concentrations to six categories from *Good* (0–50) to *Severe* (401–500) [2]. Because the AQI is published for the day that has already passed, vulnerable groups such as children, the elderly and people with asthma receive little advance warning.

Forecasting tomorrow's AQI is therefore a socially useful ML problem linked to SDG 3 (Good Health and Well-being) and SDG 11 (Sustainable Cities and Communities). Many recent studies use complex ensembles or deep networks; we ask whether the classical algorithms of an undergraduate syllabus are already sufficient when evaluated honestly. Our contributions are: (i) a leakage-free, chronological evaluation protocol with forward-chaining cross-validation; (ii) a comparison of eight regression and seven classification models against a persistence baseline; (iii) an analysis of preprocessing choices (outlier treatment, lasso feature selection and PCA); and (iv) a robustness check on the COVID-19 lockdown period.

Section 2 reviews related work, Section 3 describes the data, Sections 4–5 the methods and set-up, Section 6 the results and Section 7 the discussion, followed by conclusions.

2 Related Work

Kumar and Pande [3] analysed six years of pollution data from 23 Indian cities and compared five ML models for AQI prediction, finding that a gradient-boosting model gave the best fit between predicted and observed values. Natarajan et al. [4] combined Grey Wolf Optimisation with a decision tree to predict AQI for major Indian cities using Kaggle data and reported city-wise accuracies between about 89% and 98%. Both studies predict the AQI of the *same* day from same-day concentrations, which is useful for gap-filling but not for advance warning; we instead forecast the next day.

Mahato et al. [5] showed that the 2020 COVID-19 lockdown sharply reduced pollutant levels in Delhi, which creates a natural distribution shift that we exploit as a stress test. For evaluating predictors on time-ordered data, Bergmeir and Benítez [6] discuss when standard and blocked cross-validation are valid; we use forward-chaining splits to avoid training on the future. All models were implemented with scikit-learn [7]; the algorithms follow [8].

3 Dataset and Preprocessing

3.1 Source and description

We used *city_day.csv* from the Kaggle dataset "Air Quality Data in India (2015–2020)" [9], which compiles data from the CPCB Central Control Room portal (see Data availability). The file has 29,531 rows and 16 columns: city, date, twelve pollutant concentrations (PM2.5, PM10, NO, NO_2, NO_x, NH_3, CO, SO_2, O_3, benzene, toluene, xylene), AQI and AQI category, for 26 cities from 2015-01-01 to 2020-07-01. (Copy used: MD5 4e241e969834…). Table 1 summarises the variables. Concentrations are 24-hour city averages in the units used by CPCB.

Table 1. Variables in city_day.csv, missing values and their use in this study.

Variable	Unit	Missing %	Use
City	categorical	0.0	one-hot feature
Date	day	0.0	split, season (sin/cos)
PM2.5	µg/m^3	15.6	feature (day t)
PM10	µg/m^3	37.7	feature (day t)
NO	µg/m^3	12.1	feature (day t)
NO_2	µg/m^3	12.1	feature (day t)
NO_x	µg/m^3	14.2	feature (day t)
NH_3	µg/m^3	35.0	feature (day t)
CO	mg/m^3	7.0	feature (day t)
SO_2	µg/m^3	13.1	feature (day t)
O_3	µg/m^3	13.6	feature (day t)
Benzene	µg/m^3	19.0	feature (day t)
Toluene	µg/m^3	27.2	feature (day t)
Xylene	µg/m^3	61.3	dropped (> 50% missing)
AQI	index 0–500	15.9	feature (t, t−1, 3-day mean); target (t+1)
AQI_Bucket	6 categories	15.9	not used (derived from AQI)

3.2 Missing values

Missingness is substantial (Fig. 1): xylene is missing in 61.3% of rows and was dropped (> 50% rule); PM10 (37.7%) and NH_3 (35.0%) were retained. Remaining gaps were filled with the *training-set* median inside the model pipeline, so test data never influence the imputation. Rows without a valid AQI on day *t* or *t*+1 were removed.

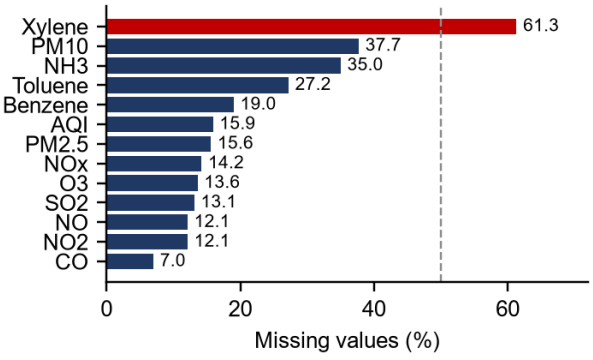


Fig. 1. Percentage of missing values per variable in city_day.csv (red: above the 50% threshold; dropped).

3.3 Feature construction, outliers and scaling

For each city the data were sorted by date and the target was defined as the AQI of the next calendar day; pairs separated by a gap were discarded. Features for day *t* are the 11 pollutant concentrations, AQI_t, AQI_{t-1}, the three-day mean AQI, the month encoded as sin/cos (to capture the winter peak in Fig. 2) and the city (one-hot, 26 levels).

The AQI scale ends at 500, but 543 records exceeded it; these were capped at 500. The IQR rule flagged between 3.2% and 10.5% of values per pollutant as outliers. Because high-pollution episodes are real events rather than errors, we winsorised each feature at its 1st and 99th training percentiles instead of deleting rows. Features were then standardised, $z = (x - \mu)/\sigma$, with µ and σ estimated on the training fold only.

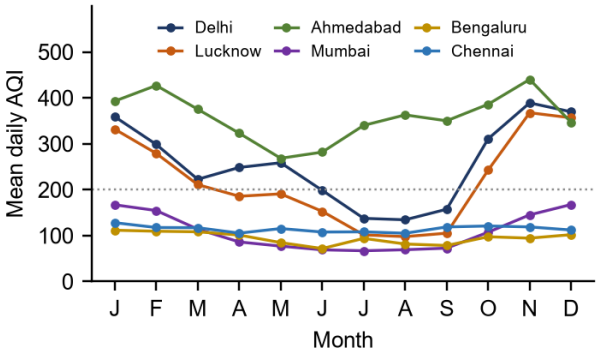


Fig. 2. Mean daily AQI by month for six cities (2015–2020). The dotted line marks AQI = 200, the threshold of the "Poor" category used as the health-alert label.

3.4 Feature selection and PCA

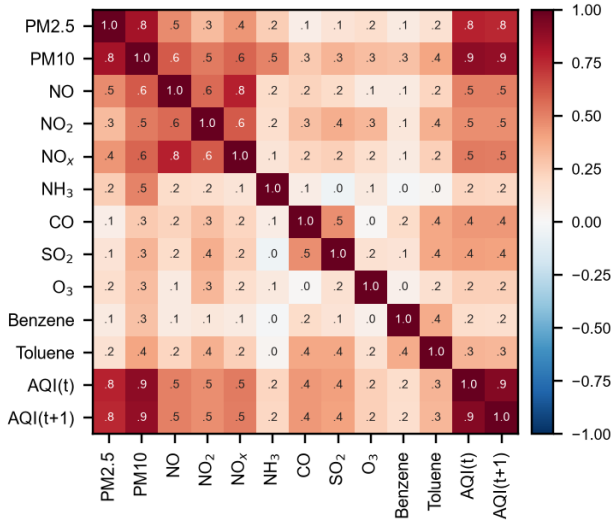


Fig. 3. Pearson correlation matrix of pollutants, current AQI and next-day AQI (training years 2015–2018, pairwise-complete observations).

Pearson correlation with next-day AQI was highest for AQI_t (0.92), the three-day mean (0.90) and PM10 (0.88). Several features are strongly inter-correlated (PM2.5–PM10 r = 0.84; NO–NO_x r = 0.78), which motivates regularisation. The tuned lasso set 7 coefficients exactly to zero (NO, NH3, Benzene, Toluene, AQI lag1, month sin,

month cos), an embedded form of feature selection. PCA on the 16 standardised numeric features (Fig. 4) needed 9 components for 90% and 12 for 95% of the variance; the first component alone explained 39.0%.

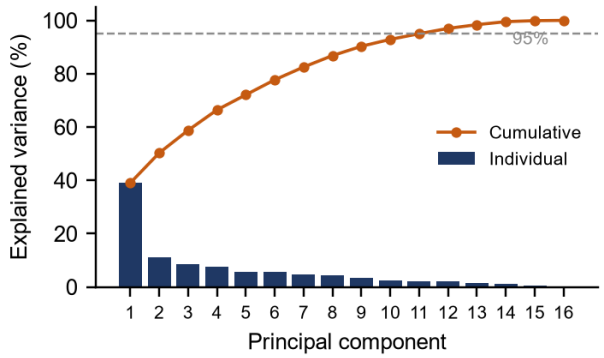


Fig. 4. Individual and cumulative explained variance of the principal components of the numeric features (training data).

4 Methodology

Fig. 5 summarises the end-to-end workflow. All steps after the chronological split are wrapped in a single pipeline that is fitted on training data only.

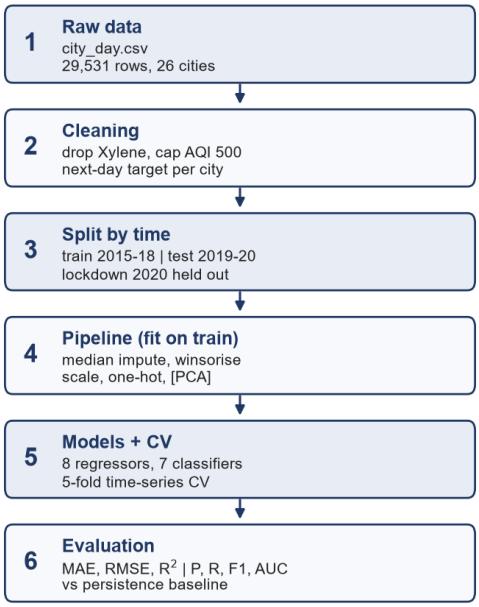


Fig. 5. Methodology pipeline used in this study (square brackets: optional step).

Let **x** be the standardised feature vector of a city-day and y the next-day AQI. The naive **persistence** baseline predicts that tomorrow equals today:

$$\hat{y}_{t+1} = AQI_t \tag{1}$$

4.1 Regression models

Multiple linear regression fits $\hat{y} = \mathbf{w}^{\mathsf{T}}\mathbf{x} + b$ by minimising the mean squared error

$$J(\mathbf{w}) = (1/n)\,\Sigma_i\,(y_i - \hat{y}_i)^2, \quad \mathbf{w} \leftarrow \mathbf{w} - \eta\,\partial J/\partial \mathbf{w} \tag{2}$$

where the second expression is the gradient-descent update with learning rate η (scikit-learn uses the closed-form least-squares solution, which reaches the same minimum). **Ridge** and **lasso** add L2 and L1 penalties to control over-fitting and multicollinearity:

$$J_{ridge} = J + \lambda\,\Sigma_j\,w_j^2, \quad J_{lasso} = J + \lambda\,\Sigma_j\,|w_j| \tag{3}$$

Polynomial regression expands the features to all degree-2 terms (squares and pairwise products) followed by ridge. A **regression tree** recursively splits the feature space to minimise within-node squared error; **k-NN regression** averages the targets of the k nearest training points under Euclidean distance $d(\mathbf{x}, \mathbf{x}') = \sqrt{\Sigma_j (x_j - x'_j)^2}$, optionally weighted by 1/d. **Support-vector regression** (SVR) fits a function that ignores errors smaller than ε and penalises larger ones linearly:

$$\min \tfrac{1}{2}\|\mathbf{w}\|^2 + C\,\Sigma_i(\xi_i + \xi_i^*), \quad |y_i - f(\mathbf{x}_i)| \le \varepsilon + \xi_i \tag{4}$$

with the RBF kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$. The **multilayer perceptron** (MLP) has two hidden layers (64 and 32 ReLU units), trained by back-propagation with the Adam optimiser, L2 penalty α and early stopping.

4.2 Classification models

The health-alert label is $c = 1$ if $AQI_{t+1} > 200$ (Poor, Very Poor or Severe) and 0 otherwise. **Logistic regression** models

$$P(c = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b), \quad \sigma(z) = 1/(1 + e^{-z}) \tag{5}$$

and is trained by minimising the regularised cross-entropy (inverse strength C). **Gaussian Naive Bayes** assumes conditionally independent, normally distributed features: $P(c \mid \mathbf{x})$ is proportional to $P(c)\,\Pi_j\,N(x_j; \mu_{jc}, \sigma^2_{jc})$. The **decision tree** chooses splits that minimise Gini impurity $1 - \Sigma_k\,p_k^2$. **k-NN** takes a majority vote of the k nearest neighbours. The single-layer **perceptron** updates $\mathbf{w} \leftarrow \mathbf{w} + \eta(c_i - \hat{c}_i)\mathbf{x}_i$ on each mistake. The soft-margin **SVM** with RBF kernel solves the classification analogue of (4), $y_i(\mathbf{w}^T\phi(\mathbf{x}_i) + b) \ge 1 - \xi_i$, and the **MLP** classifier mirrors the regressor with a logistic output. Class imbalance (about one alert day in three in training) was handled by treating class_weight = "balanced" as a hyper-parameter.

4.3 Evaluation metrics

Regression: $MAE = (1/n)\Sigma|y - \hat{y}|$, $RMSE = \sqrt{[(1/n)\Sigma(y - \hat{y})^2]}$ and $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$. Classification: accuracy, precision = TP/(TP+FP), recall = TP/(TP+FN), F1 = 2PR/(P+R) and the area under the ROC curve (AUC). Because a missed alert (false negative) is costlier for public health than a false alarm, recall and F1 are emphasised over accuracy.

5 Experimental Setup

Split. To mimic real forecasting, data were split by time, not at random: training = 2015–2018 (13,041 samples), test = 1 January 2019 to 29 February 2020 (8,374). The national lockdown period, 25 March to 30 June 2020 (2,369), was held out as a separate stress test. The alert rate was 33.1% in training and 23.2% in the test set.

Cross-validation and tuning. Hyper-parameters were chosen by grid search with five-fold *forward-chaining* cross-validation (TimeSeriesSplit) on the training years, optimising RMSE for regression and F1 for classification. All preprocessing (imputation, winsorisation, scaling,

encoding, PCA) was placed inside a scikit-learn Pipeline so that it was re-fitted on each training fold, preventing data leakage. Table 2 lists the grids and the selected values.

Unseen cities. 7 cities (Guwahati, Coimbatore, Chandigarh, Bhopal, Shillong, Kochi, Ernakulam) have data only after 2018 and therefore appear only in the test set; their one-hot city indicator is all-zero (handle_unknown = "ignore"), so the model must generalise from other cities.

Environment. Python 3.12, scikit-learn 1.9, pandas, NumPy and Matplotlib on a Windows 11 laptop; random seed 42 for all stochastic models. The full code (run_experiments.py) regenerates every number and figure in this paper.

Table 2. Hyper-parameter grids and values selected by 5-fold forward-chaining CV (cw = class weight).

Model	Grid searched	Selected
Ridge / Lasso	λ in {0.1…100} / {0.01…5}	alpha=100 / alpha=1
Poly-2 + Ridge	λ in {10, 100, 1000}	alpha=100
Tree (reg.)	depth {4,6,8,12}; min leaf {5,20,50}	depth=6, min leaf=50
k-NN (reg.)	k {5,15,31}; uniform/distance	k=31, weights=distance
SVR (RBF)	C {10,100}; ε {5,15}	C=100, ε=15
MLP (reg./clf.)	α {10^{-4}, 10^{-2}, 1}	α=0.0001 / α=1
Logistic reg.	C {0.01…10}; class weight	C=1, cw=None
Tree (clf.)	depth {4…12}; class weight	cw=balanced, depth=4
k-NN (clf.)	k {5,15,31}	k=31
SVM (RBF)	C {0.1,1,10}; class weight	C=1, cw=None

6 Results

6.1 Next-day AQI regression

Table 3. Next-day AQI regression on the 2019–20 test set (AQI units). CV-RMSE is the mean over the five training folds. Best test RMSE in bold.

Model	MAE	RMSE	R^2	CV-RMSE
Persistence baseline	25.59	42.43	0.839	–
Linear reg. (OLS)	24.55	36.90	0.879	43.04
Ridge reg.	24.45	36.98	0.878	42.19
Lasso reg.	24.39	37.75	0.873	42.26
Polynomial (deg 2) + Ridge	24.37	35.26	0.889	41.75
Decision tree	23.33	36.39	0.882	44.15
k-NN regressor	28.15	40.39	0.855	46.29
SVR (RBF)	**23.69**	**34.96**	**0.891**	**44.41**
MLP (64-32)	26.08	38.46	0.868	43.51

Table 3 shows that every learned model beats persistence (RMSE 42.43). The SVR achieved the lowest RMSE (34.96, a 17.6% reduction) and highest R^2 (0.891), followed closely by the degree-2 polynomial ridge model (35.26). The shallow regression tree had the lowest MAE (23.33) but a higher RMSE, i.e. it is accurate on typical days but makes larger errors on extreme days. Plain OLS (36.90) was within 5.5% of the best model, so most of the predictable signal is linear in today's AQI and particulate levels. k-NN was the weakest learned model, which is expected with 40+ dimensions after one-hot encoding. Test errors were lower than CV errors, probably because the test period was less polluted (alert rate 23.2% vs 33.1% in training) and large AQI values produce large absolute errors.

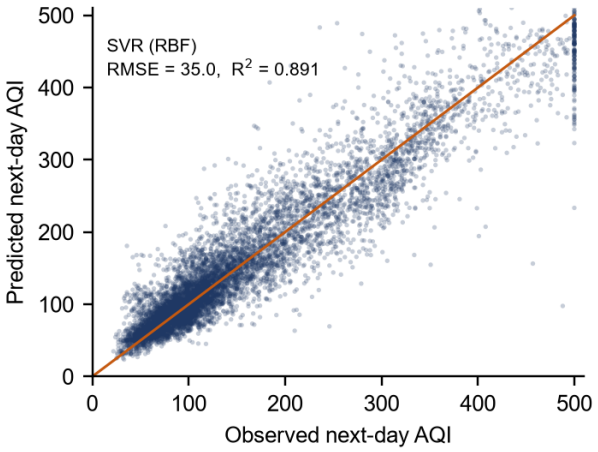


Fig. 6. Observed versus predicted next-day AQI for the best regressor (SVR (RBF)) on the test set; the orange line is perfect prediction.

Fig. 6 shows close agreement over most of the range. The vertical band at 500 contains days whose AQI was capped at the top of the scale; many of these extreme days are under-predicted, while very clean days (AQI < 60) are slightly over-predicted – a typical regression-to-the-mean effect. The lasso coefficients (standardised scale) ranked AQI_t, PM2.5, the three-day mean and CO as the strongest predictors.

6.2 Health-alert classification

Table 4. Health-alert (next-day AQI > 200) classification on the 2019–20 test set. Best F1 in bold.

Model	Acc.	Prec.	Recall	F1	AUC
Persistence baseline	0.922	0.832	0.834	0.833	–
Logistic regression	**0.938**	**0.873**	**0.860**	**0.866**	**0.981**
Gauss. Naive Bayes	0.865	0.656	0.882	0.752	0.935
Decision tree	0.927	0.791	0.931	0.855	0.974
k-NN	0.924	0.857	0.807	0.831	0.970
Perceptron	0.921	0.855	0.794	0.823	0.966
SVM (RBF)	0.938	0.870	0.863	0.866	0.979
MLP (64-32)	0.937	0.865	0.865	0.865	0.981

Logistic regression, the RBF-SVM and the MLP were practically tied (F1 0.865–0.866, AUC 0.979–0.981), each improving on persistence (F1 0.833) mainly through higher precision. The depth-4 tree with balanced class weights had the highest recall (0.931) at the cost of more false alarms, which may be preferable for a cautious public-health agency. Gaussian Naive Bayes had the lowest F1 (0.752) because its independence assumption is badly violated by the correlated pollutants. For logistic regression the confusion matrix was TN = 6,187, FP = 243, FN = 273, TP = 1,671: about one alert day in seven was missed.

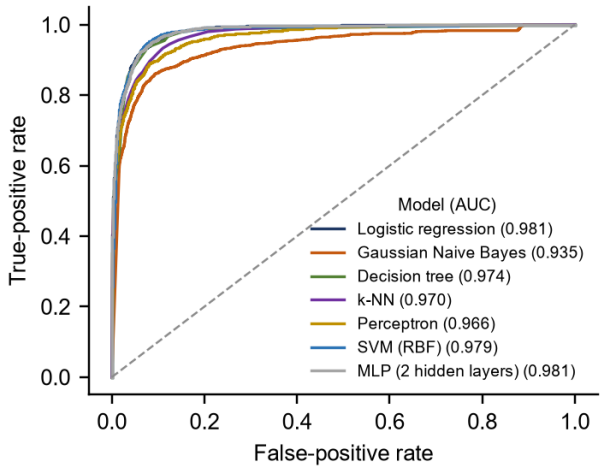


Fig. 7. ROC curves of the seven classifiers on the test set (AUC in brackets).

6.3 Effect of PCA and robustness to the lockdown

Table 5. Test RMSE with all features vs PCA (95% variance), and next-day AQI regression during the 2020 lockdown.

Setting	MAE	RMSE	R^2
OLS, all features	24.55	36.90	0.879
OLS, PCA 95%	24.51	37.57	0.874
k-NN (k=15), all features	28.38	40.64	0.853
k-NN (k=15), PCA 95%	27.66	40.10	0.857
Lockdown: persistence	11.53	21.11	0.734
Lockdown: SVR (RBF)	16.91	25.18	0.621

PCA slightly hurt the linear model but slightly helped k-NN, whose distance computations benefit from removing redundant, correlated directions. In the lockdown period the tuned SVR was clearly *worse* than persistence: emissions fell abruptly, so relationships learned from 2015–2018 over-predicted pollution, while persistence adapts automatically.

7 Discussion

What worked. A small set of features (today's AQI, particulate levels, season and city) carries most of the information; regularised linear and kernel models capture it with little tuning and remain interpretable. For a city administration, logistic regression offers nearly the best F1 together with coefficients that can be explained to the public.

Limitations. (i) The data stop in July 2020 and contain no meteorological variables (wind, temperature, humidity, boundary-layer height), which drive day-to-day changes; adding IMD or reanalysis weather would likely help. (ii) City-level averages hide large differences between monitoring stations. (iii) Heavy missingness was handled by simple median imputation. (iv) The lockdown result shows that models must be monitored and re-trained when emission patterns change; a fixed model should never replace expert judgement.

Performance across cities. Per-city errors (Appendix A, Table A1) reveal an equity issue hidden by the overall score. The SVR beat persistence in 20 of 25 cities, but was worse in 5 (Kolkata, Coimbatore, Shillong, Kochi, Ernakulam), almost all of which had little or no training data. Smaller cities would therefore receive less reliable forecasts unless city-specific data are collected or a fallback to persistence is used.

Ethics and SDG impact. The data are aggregate environmental measurements with no personal information. The main ethical risk is over-reliance: a missed alert could expose vulnerable people, so a deployed system should favour recall, publish its uncertainty and be audited across cities, since smaller cities with fewer monitors may receive less accurate forecasts. Reliable next-day alerts contribute to SDG target 3.9 (reducing deaths and illnesses from air pollution) and SDG target 11.6 (reducing the adverse environmental impact of cities, including air quality).

8 Conclusion and Future Work

Using real CPCB-derived data for 26 Indian cities and a leakage-free chronological protocol, we showed that syllabus-level ML models forecast next-day AQI with $R^2 \approx 0.89$ and issue "Poor"-day alerts with F1 ≈ 0.87, clearly beating a persistence baseline in normal conditions. Future work will add weather forecasts, extend the data beyond 2020 from the CPCB portal, use station-level data and test online re-training to handle sudden changes such as lockdowns or crop-residue burning seasons.

Declarations

Data availability. The dataset is publicly available at https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india (accessed 06 Oct 2026); the original source is the CPCB Central Control Room. Code and figures: folder *sample_paper/* of this assignment pack (a student places the code on GitHub).

Conflict of interest. The author declares no conflict of interest.
Funding. None.

Use of AI tools. This SAMPLE was drafted by an AI assistant (Claude, Anthropic) for the course instructor to illustrate the expected format; the experiments were actually executed with the accompanying code and every number is produced by it. In a real submission, the student must state precisely which AI tools were used and for what (e.g. grammar checking, debugging), and confirm that all content was verified and that they take full responsibility for it.

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Appendix A. Per-city test error

Table A1. Next-day AQI RMSE by city on the 2019–20 test set for the selected SVR and for persistence (overall SVR RMSE = 34.96). n = number of samples; cities with $n_{train} = 0$ were unseen in training.

City	n_{train}	n_{test}	Mean AQI	RMSE SVR	RMSE pers.
Bengaluru	1354	425	92	12.7	13.2
Chennai	1325	425	100	28.5	33.2
Delhi	1446	425	238	38.1	51.3
Hyderabad	1320	425	95	13.2	14.6
Jaipur	538	425	120	27.6	36.1
Kolkata	**205**	**425**	**152**	**40.3**	**35.9**
Lucknow	1332	425	208	33.6	49.1
Mumbai	222	425	116	21.3	23.0
Gurugram	856	423	198	41.4	53.7
Amritsar	566	416	110	30.6	39.0
Patna	872	415	220	39.9	43.8
Thiruvananthapuram	511	407	77	16.4	18.5
Ahmedabad	780	402	404	70.6	90.5
Visakhapatnam	625	382	122	23.8	32.3
Guwahati	0	368	150	51.3	66.8
Amaravati	313	363	95	20.5	23.8
Brajrajnagar	264	352	149	32.7	41.2
Jorapokhar	283	351	159	38.8	41.8
Talcher	229	341	181	38.9	47.1
Coimbatore	**0**	**235**	**78**	**20.7**	**9.5**
Chandigarh	0	174	122	32.1	39.9
Bhopal	0	165	157	31.7	42.5
Shillong	**0**	**106**	**61**	**33.3**	**16.1**
Kochi	**0**	**38**	**130**	**46.3**	**34.5**
Ernakulam	**0**	**36**	**102**	**65.8**	**15.4**

Rows in bold: cities where the model was worse than persistence. Generated by per_city.py.

SAMPLE